Solution Architecture

Date	23 June 2026
Team ID	XXX
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:
Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Presentation / Client Layer
HTML, CSS, JavaScript

↓ ↑

Web Application / Backend
Flask (Python)

↓ ↑

Input & Validation Service N, P, K, Temperature Humidity, pH, Rainfall	Core Logic Service OptiCrop Prediction Engine Pre-trained ML Model Crop Recommendation	Model Integration scikit-learn Pickle Model (crop_model.pkl)

↓ ↑

Data / Storage Layer
Crop Dataset, Trained Model, Prediction Inputs, GitHub Repository

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
[Presentation Layer]	Provides the interface for entering agricultural parameters and viewing crop recommendations.	HTML, CSS, JavaScript
[Backend / Web Layer]	Handles requests, validates input, loads the trained model and returns prediction results.	Python, Flask
[ML Prediction Engine]	Uses N, P, K, temperature, humidity, pH and rainfall to recommend a suitable crop.	scikit-learn, Pickle
[Data / Model Storage]	Stores the crop dataset, serialized trained model and project files.	CSV Dataset, crop_model.pkl, GitHub